

21.5 Serial correlation and the Durbin–Watson test

When observations are taken in sequence, over time or along a transect, neighbouring errors are often alike. A shock to the economy lasts more than one quarter, a drifting instrument affects consecutive readings, and a variable missing from the model usually changes slowly, so its effect appears in the errors as a slow wave. Errors that are correlated with their neighbours are **serially correlated** or **autocorrelated**. As with heteroscedasticity, least squares stays unbiased and its usual standard errors become wrong (Theorem 19.2.2). The error is typically in the dangerous direction: with positive autocorrelation and a slowly varying regressor, the true variance of a slope can be many times its nominal value. [Section 21.6](#) measures this. This section is about detection.

21.5.1 Looking at residuals in time order

Three displays are standard. The residuals plotted against time show runs of the same sign when the autocorrelation is positive, and rapid alternation when it is negative. The **lag plot** of $\hat{\varepsilon}_t$ against $\hat{\varepsilon}_{t-1}$ shows the lag-one dependence as a tilt. And the **residual autocorrelations**

$$r_k = \frac{\sum_{t=k+1}^n \hat{\varepsilon}_t \hat{\varepsilon}_{t-k}}{\sum_{t=1}^n \hat{\varepsilon}_t^2}, \quad k = 1, 2, \dots,$$

summarize dependence at each lag. Under the model the residuals are themselves slightly correlated, with covariance $\sigma^2(\mathbf{I} - \mathbf{M})$, so the r_k are not exactly centred at zero, but for large n they are approximately independent with standard deviation $n^{-1/2}$.

Example. A quarterly Okun’s-law regression

Okun’s law, named after the economist Arthur Okun, relates changes in unemployment to output growth. The statsmodels macroeconomic data give US quarterly series from 1959 to 2009. Regress the change in the unemployment rate (percentage points per quarter) on real GDP growth (percent, at an annual rate) for the $n = 202$ quarters from the second quarter of 1959. The fitted line is

$$\widehat{\Delta u}_t = 0.227 - 0.0671 g_t,$$

so unemployment is stable when output grows at about $0.227/0.0671 \approx 3.4$ percent a year, and each extra percentage point of growth lowers it by about 0.07 points per quarter. [Figure 21.5.1\(a,b\)](#) shows the residuals. They wander in long swings, and the lag plot tilts, with $r_1 = 0.300$.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.integrate import quad
from scipy.optimize import brentq

macro = sm.datasets.macrodta.load_pandas().data
# change in unemployment rate
du = np.diff(macro["unemp"].to_numpy())
# GDP growth, % a year
growth = 400 * np.diff(np.log(macro["realgdp"].to_numpy()))
year = (macro["year"] + (macro["quarter"] - 1) / 4).to_numpy()[1:]
```

```

n = len(du)
X = np.column_stack([np.ones(n), growth])
beta, *_ = np.linalg.lstsq(X, du, rcond=None)
e = du - X @ beta
# Durbin-Watson statistic
d = np.sum(np.diff(e) ** 2) / np.sum(e**2)
# lag-one residual autocorrelation
r1 = np.sum(e[1:] * e[:-1]) / np.sum(e**2)
print(f"n = {n}, slope {beta[1]:.4f}, d = {d:.3f}, r1 = {r1:.3f}, "
      f"2(1 - r1) = {2 * (1 - r1):.3f}")

```

[Run this code online](#)

21.5.2 The Durbin–Watson statistic

Definition 21.5.1. Durbin–Watson statistic

For residuals $\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n$ in time order,

$$d = \frac{\sum_{t=2}^n (\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2}{\sum_{t=1}^n \hat{\varepsilon}_t^2} = \frac{\hat{\varepsilon}^\top \mathbf{A} \hat{\varepsilon}}{\hat{\varepsilon}^\top \hat{\varepsilon}}, \quad \mathbf{A} = \mathbf{D}^\top \mathbf{D},$$

where $\mathbf{D}$ is the $(n-1) \times n$ first-difference matrix, with rows $(\dots, 0, -1, 1, 0, \dots)$.

Expanding the numerator gives $d = 2(1 - r_1) - (\hat{\varepsilon}_1^2 + \hat{\varepsilon}_n^2) / \sum_t \hat{\varepsilon}_t^2$, so $d \approx 2(1 - r_1)$. Values near 2 indicate no lag-one correlation, values towards 0 positive correlation, and values towards 4 negative correlation. For the Okun regression $d = 1.386$, against $2(1 - r_1) = 1.401$.

The matrix $\mathbf{A}$ is tridiagonal with diagonal $(1, 2, \dots, 2, 1)$ and off-diagonal entries -1 . Its spectrum is known exactly, and that is what makes the Durbin–Watson theory work.

Lemma 21.5.1. Eigenvalues of the difference matrix

The eigenvalues of $\mathbf{A}$ are

$$\lambda_j = 4 \sin^2\left(\frac{\pi(j-1)}{2n}\right), \quad j = 1, \dots, n,$$

with eigenvectors $\mathbf{v}_j = (\cos(\theta_j/2), \cos(3\theta_j/2), \dots, \cos((2n-1)\theta_j/2))^\top$, where $\theta_j = \pi(j-1)/n$. In particular $0 = \lambda_1 < \lambda_2 < \dots < \lambda_n < 4$, and $\mathbf{v}_1 = \mathbf{1}$.

Proof. Let $\theta = \theta_j$ and $v_t = \cos((t - \frac{1}{2})\theta)$, and extend the sequence by $v_0 = \cos(-\theta/2) = v_1$ and $v_{n+1} = \cos((n + \frac{1}{2})\theta)$. Since $n\theta = \pi(j-1)$, $v_{n+1} = \cos(\pi(j-1) + \theta/2) = \cos(\pi(j-1) - \theta/2) = v_n$. With these boundary values every row of $\mathbf{A}\mathbf{v}$ has the interior form $(\mathbf{A}\mathbf{v})_t = 2v_t - v_{t-1} - v_{t+1}$, because the first row $v_1 - v_2$ equals $2v_1 - v_0 - v_2$ and similarly for the last. By the identity $\cos(a-\theta) + \cos(a+\theta) = 2\cos a \cos \theta$, $2v_t - v_{t-1} - v_{t+1} = 2(1 - \cos \theta)v_t = 4\sin^2(\theta/2)v_t$. The n values $\theta_j \in [0, \pi)$ give n distinct eigenvalues, so these are all of them. ■

The distribution theory also needs a classical fact about eigenvalues of compressed matrices.

Lemma 21.5.2. Interlacing

Let $\mathbf{B}$ be $N \times N$ symmetric with eigenvalues $\beta_1 \leq \dots \leq \beta_N$, and let $\mathbf{U}$ be $N \times m$ with orthonormal columns. If $\mu_1 \leq \dots \leq \mu_m$ are the eigenvalues of $\mathbf{U}^\top \mathbf{B} \mathbf{U}$, then $\beta_k \leq \mu_k \leq \beta_{k+N-m}$ for $k = 1, \dots, m$.

Proof. Let $\mathbf{f}_1, \dots, \mathbf{f}_m$ be orthonormal eigenvectors of $\mathbf{U}^\top \mathbf{B} \mathbf{U}$ and $\mathbf{q}_1, \dots, \mathbf{q}_N$ those of $\mathbf{B}$. The subspaces $\mathcal{S} = \text{span}\{\mathbf{U}\mathbf{f}_1, \dots, \mathbf{U}\mathbf{f}_m\}$, of dimension m , and $\mathcal{T} = \text{span}\{\mathbf{q}_k, \dots, \mathbf{q}_N\}$, of dimension $N - k + 1$, have dimensions adding to $N + 1$, so they share a nonzero vector $\mathbf{x} = \mathbf{U}\mathbf{f}$. Since $\mathbf{U}$ preserves lengths, $\mathbf{x}^\top \mathbf{B} \mathbf{x} / \mathbf{x}^\top \mathbf{x} = \mathbf{f}^\top \mathbf{U}^\top \mathbf{B} \mathbf{U} \mathbf{f} / \mathbf{f}^\top \mathbf{f} \leq \mu_k$, because $\mathbf{f} \in \text{span}\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$; and $\mathbf{x}^\top \mathbf{B} \mathbf{x} / \mathbf{x}^\top \mathbf{x} \geq \beta_k$, because $\mathbf{x} \in \mathcal{T}$. Both bounds are Theorem 1.7.7 applied to the restriction of the quadratic form to the span of eigenvectors. So $\beta_k \leq \mu_k$. The upper bound follows in the same way from $\text{span}\{\mathbf{U}\mathbf{f}_k, \dots, \mathbf{U}\mathbf{f}_m\}$ and $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_{k+N-m}\}$, whose dimensions add to $N + 1$. ■

21.5.3 The null distribution of d **Theorem 21.5.3. Durbin–Watson**

Let $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ with $\text{rank}(\mathbf{X}) = p < n$, let the columns of the $n \times (n - p)$ matrix $\mathbf{Q}_\perp$ be an orthonormal basis of $\mathcal{C}(\mathbf{X})^\perp$, and let $\nu_1 \leq \dots \leq \nu_{n-p}$ be the eigenvalues of $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp$. Put $m = n - p$.

- (a) d has the distribution of $\sum_k \nu_k \xi_k^2 / \sum_k \xi_k^2$, where $\xi_1, \dots, \xi_m$ are independent $N(0, 1)$. It depends on $\mathbf{X}$ but not on $\boldsymbol{\beta}$ or σ^2 , and d is independent of $\text{SSE} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}}$.
- (b) $E d = \bar{\nu} = m^{-1} \sum_k \nu_k = \text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{A}\} / m$ and

$$\text{Var } d = \frac{2 \sum_k (\nu_k - \bar{\nu})^2}{m(m + 2)}.$$

- (c) $\Pr(d \leq c) = \Pr\{\sum_k (\nu_k - c) \xi_k^2 \leq 0\}$ for every c .
- (d) (*Bounds.*) If $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, then $\lambda_{k+1} \leq \nu_k \leq \lambda_{k+p}$ for $k = 1, \dots, m$, with λ_j as in Lemma 21.5.1. Consequently $d_L \leq d \leq d_U$ with probability one, where, for the same ξ_k ,

$$d_L = \frac{\sum_k \lambda_{k+1} \xi_k^2}{\sum_k \xi_k^2}, \quad d_U = \frac{\sum_k \lambda_{k+p} \xi_k^2}{\sum_k \xi_k^2},$$

whose distributions depend only on n and p .

Proof.

- (a) $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon} = \mathbf{Q}_\perp \mathbf{Q}_\perp^\top \boldsymbol{\varepsilon}$, because $\mathbf{Q}_\perp \mathbf{Q}_\perp^\top$ is the projection onto $\mathcal{C}(\mathbf{X})^\perp$. Let $\boldsymbol{\zeta} = \mathbf{Q}_\perp^\top \boldsymbol{\varepsilon} / \sigma \sim N_m(\mathbf{0}, \mathbf{I}_m)$ (Theorem 3.3.1), and take a spectral decomposition $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp = \mathbf{P} \text{diag}(\nu_k) \mathbf{P}^\top$ with $\mathbf{P}$ orthogonal. Then $\boldsymbol{\xi} = \mathbf{P}^\top \boldsymbol{\zeta} \sim N_m(\mathbf{0}, \mathbf{I}_m)$, $\hat{\boldsymbol{\varepsilon}}^\top \mathbf{A} \hat{\boldsymbol{\varepsilon}} = \sigma^2 \boldsymbol{\zeta}^\top \mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp \boldsymbol{\zeta} = \sigma^2 \sum_k \nu_k \xi_k^2$ and $\hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \sigma^2 \|\boldsymbol{\zeta}\|^2 = \sigma^2 \sum_k \xi_k^2$. For independence, put $G_k = \xi_k^2 / 2$, which are independent gamma variables with shape $\frac{1}{2}$ and

density proportional to $g^{-1/2}e^{-g}$. Change variables from $(G_1, \dots, G_m)$ to $S = \sum_k G_k$ and $W_k = G_k/S$, $k < m$; the Jacobian is S^{m-1} , and the joint density becomes proportional to $S^{m/2-1}e^{-S} \prod_k w_k^{-1/2}$ (with $w_m = 1 - \sum_{k < m} w_k$), a product of a function of S and a function of the W_k . So $(W_1, \dots, W_m)$ is independent of S . Now $d = \sum_k \nu_k W_k$ and $\text{SSE} = 2\sigma^2 S$.

- (b) Let $N = \sum_k \nu_k \xi_k^2$ and $S_2 = \sum_k \xi_k^2$, so $N = dS_2$ with d independent of S_2 . Then $\mathbb{E} N = \mathbb{E} d \mathbb{E} S_2$ and $\mathbb{E} N^2 = \mathbb{E} d^2 \mathbb{E} S_2^2$. Here $\mathbb{E} N = \sum_k \nu_k$, $\mathbb{E} S_2 = m$, $\mathbb{E} N^2 = \text{Var } N + (\mathbb{E} N)^2 = 2 \sum_k \nu_k^2 + (\sum_k \nu_k)^2$ and $\mathbb{E} S_2^2 = 2m + m^2$. Dividing gives $\mathbb{E} d = \bar{\nu}$ and $\mathbb{E} d^2 = \{2 \sum_k \nu_k^2 + (\sum_k \nu_k)^2\} / \{m(m+2)\}$, and $\text{Var } d = \mathbb{E} d^2 - \bar{\nu}^2$ simplifies to the stated form. Finally $\sum_k \nu_k = \text{tr}(\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp) = \text{tr}(\mathbf{A} \mathbf{Q}_\perp \mathbf{Q}_\perp^\top) = \text{tr}\{\mathbf{A}(\mathbf{I} - \mathbf{M})\}$.
- (c) $d \leq c$ iff $\sum_k \nu_k \xi_k^2 \leq c \sum_k \xi_k^2$.
- (d) Let $\mathbf{P}_1$ be the $n \times (n-1)$ matrix whose columns are the normalized eigenvectors $\mathbf{v}_2, \dots, \mathbf{v}_n$ of $\mathbf{A}$; they span $\mathbf{1}^\perp$. Since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $\mathcal{C}(\mathbf{Q}_\perp) \subseteq \mathbf{1}^\perp$, so $\mathbf{Q}_\perp = \mathbf{P}_1 \mathbf{U}$ for an $(n-1) \times m$ matrix $\mathbf{U} = \mathbf{P}_1^\top \mathbf{Q}_\perp$ with orthonormal columns. Then $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp = \mathbf{U}^\top \text{diag}(\lambda_2, \dots, \lambda_n) \mathbf{U}$, and [Lemma 21.5.2](#) with $N = n-1$ gives $\lambda_{k+1} \leq \nu_k \leq \lambda_{k+1+(n-1-m)} = \lambda_{k+p}$. Multiplying by ξ_k^2 , summing, and dividing by $\sum_k \xi_k^2$ gives $d_L \leq d \leq d_U$. The λ_j depend only on n . ■

Part (a) reduces the exact null distribution to that of a ratio of quadratic forms in normal variables, of the kind studied in [Chapter 4](#). Part (c) turns its distribution function into that of a single indefinite quadratic form, whose distribution function Imhof (1961) showed how to compute by numerical inversion of its characteristic function. So the exact p-value, given $\mathbf{X}$, is a routine computation, and the listing below does it. When Durbin and Watson (1950, 1951) proposed the test, that computation was impractical, and part (d) was their way around it. The bounds d_L and d_U do not depend on $\mathbf{X}$, so their percentage points can be tabulated once for each n and p . The **bounds test** of the hypothesis of no autocorrelation against positive autocorrelation, at level α , rejects if $d < d_{L,\alpha}$, accepts if $d > d_{U,\alpha}$, and is inconclusive in between. To test against negative autocorrelation, use $4 - d$.

Example. The exact test for the Okun regression

Under the hypothesis of independent normal errors, and given the growth series, d has mean 2.0031 and standard deviation 0.1399, and its 5% point is 1.773. The bounds for $n = 202$, $p = 2$ are $d_{L,0.05} = 1.760$ and $d_{U,0.05} = 1.780$, close together for a sample this large. The observed $d = 1.386$ is far below both, and the exact one-sided p-value is 3.3×10^{-6} .

The bounds matter in small samples. For the 40 quarters from the second quarter of 1999 to the first quarter of 2009, the same regression gives $d = 1.478$, between the bounds 1.442 and 1.544: the bounds test is inconclusive. The exact p-value, 0.037, settles it. [Figure 21.5.1\(c\)](#) shows the three null densities for this window.

```
def dw_matrix(n):
    """A = D'D, where D takes first differences, so that d = e'Ae / e'e."""
    D = np.diff(np.eye(n), axis=0)
    return D.T @ D

def null_eigenvalues(X):
    """The n - p eigenvalues nu_k of the DW numerator on C(X)-perp."""
```

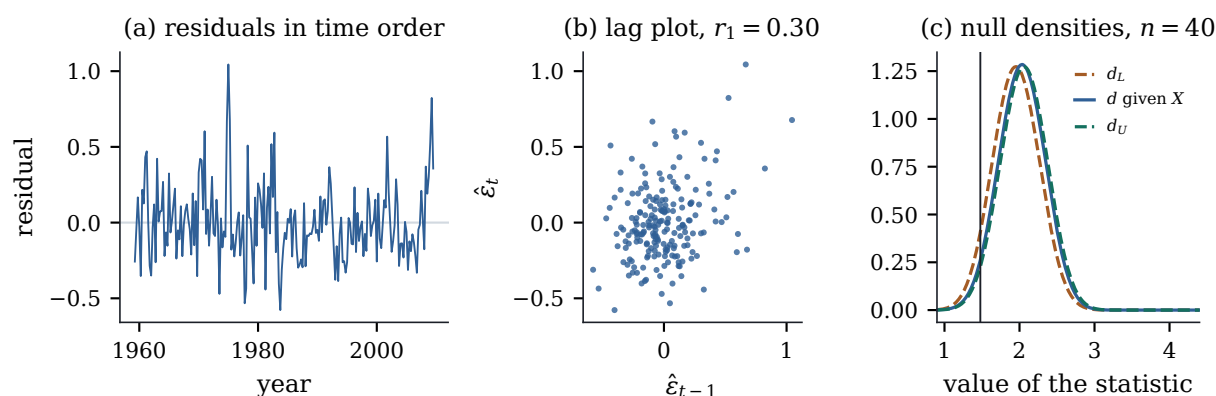


Figure 21.5.1: The Okun regression. (a) Residuals in time order. (b) Lag plot. (c) For the 40-quarter window 1999–2009: the exact null density of d given the regressors, between the densities of the bounds d_L and d_U ; the vertical line marks the observed d .

```

n, p = X.shape
Qfull, _ = np.linalg.qr(X, mode="complete")
# orthonormal basis of C(X)-perp
Qp = Qfull[:, p:]
return np.linalg.eigvalsh(Qp.T @ dw_matrix(n) @ Qp)

def prob_negative(c):
    """Pr(sum_k c_k x_{i_k}^2 < 0), x_{i_k} iid standard normal (Imhof's formula)."""
    def integrand(u):
        theta = 0.5 * np.sum(np.arctan(c * u))
        rho = np.exp(0.25 * np.sum(np.log1p((c * u) ** 2)))
        return np.sin(theta) / (u * rho)
    return 0.5 - quad(integrand, 0, np.inf, limit=400)[0] / np.pi

nu = null_eigenvalues(X)
# Pr(d <= observed) under H0
p_value = prob_negative(nu - d)
m = len(nu)
mean_d = nu.mean()
var_d = 2 * (m * np.sum(nu**2) - np.sum(nu) ** 2) / (m**2 * (m + 2))
print(f"exact one-sided p-value {p_value:.2e}; "
      f"null mean {mean_d:.4f}, sd {np.sqrt(var_d):.4f}")

```

[Run this code online](#)

```

w = slice(160, 200)
Xw, yw = X[w], du[w]
ew = yw - Xw @ np.linalg.lstsq(Xw, yw, rcond=None)[0]
d_w = np.sum(np.diff(ew) ** 2) / np.sum(ew**2)
nw_, pw_ = Xw.shape
lam_w = 4 * np.sin(np.pi * np.arange(nw_) / (2 * nw_)) ** 2
dL_w = brentq(lambda t: prob_negative(lam_w[1:nw_ - pw_ + 1] - t) - 0.05,
              0.5, 2.0)
dU_w = brentq(lambda t: prob_negative(lam_w[pw_:] - t) - 0.05, 0.5, 2.0)
p_w = prob_negative(null_eigenvalues(Xw) - d_w)
print(f"1999Q2-2009Q1: d = {d_w:.3f}, bounds ({dL_w:.3f}, {dU_w:.3f}), "
      f"exact p = {p_w:.3f}")

```

[Run this code online](#)

Three limitations should be kept in mind. The exact theory needs normal errors and regressors fixed in the sense of [Chapter 14](#): conditional on the regressors the argument goes through as long as they are not functions of past responses. It fails when a lagged response is among the regressors, and then d is biased towards 2, hiding autocorrelation; Durbin (1970) gave an alternative for that case. The bounds need an intercept. And the statistic is aimed at lag-one dependence. It is sensitive to other forms only insofar as they produce lag-one correlation, and it can miss, for example, quarterly seasonal correlation at lag four.

21.5.4 The Breusch–Godfrey test

A more flexible test comes from the score principle of [Section 21.2](#). Regress the residual $\hat{\varepsilon}_t$ on the original regressors $\mathbf{x}_{(t)}$ and on the lagged residuals $\hat{\varepsilon}_{t-1}, \dots, \hat{\varepsilon}_{t-q}$, with lagged values before the first observation set to zero, and let R^2 be the coefficient of determination. The statistic nR^2 is the score statistic for the hypothesis of no autocorrelation against autoregressive, or moving-average, errors of order q , and under the hypothesis it is approximately $\chi^2(q)$ in large samples (Breusch 1978; Godfrey 1978). Unlike d , it remains valid when lagged responses are among the regressors, and it looks at several lags at once. We state its null limit without proof; the argument parallels that of Theorem 21.2.2(c).

For the Okun regression with $q = 4$, $nR^2 = 22.43$ on four degrees of freedom, with p-value 1.6×10^{-4} . Portmanteau statistics such as that of Ljung and Box (1978), based on $\sum_{k \leq q} r_k^2$, serve the same purpose in time-series modelling.

```
lags = np.column_stack([np.concatenate([np.zeros(k), e[:-k]])
                        for k in range(1, 5)])
Z = np.column_stack([X, lags])
coef, *_ = np.linalg.lstsq(Z, e, rcond=None)
# n R^2
bg = n * (1 - np.sum((e - Z @ coef) ** 2) / np.sum((e - e.mean()) ** 2))
print(f"Breusch-Godfrey (4 lags) = {bg:.2f}, p = {stats.chi2.sf(bg, 4):.1e}")
```

[Run this code online](#)

21.5.5 Exercises

21.5.5.1 A. Check your understanding

Exercise. A1

Show that $0 \leq d < 4$ and that $d = 2(1 - r_1) - (\hat{\varepsilon}_1^2 + \hat{\varepsilon}_n^2) / \sum_t \hat{\varepsilon}_t^2$. Which residual patterns give values near 0 and near 4?

Exercise. A2

For the intercept-only model $\mathbf{X} = \mathbf{1}$, show that $E d = 2$ exactly under the hypothesis.

Solution. By [Theorem 21.5.3\(b\)](#), $E d = \text{tr}\{(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{A}\} / (n-1) = \{\text{tr } \mathbf{A} - n^{-1}\mathbf{1}^\top \mathbf{A} \mathbf{1}\} / (n-1)$. The diagonal of $\mathbf{A}$ sums to $2(n-1)$, and $\mathbf{A} \mathbf{1} = \mathbf{0}$. So $E d = 2$.

21.5.5.2 B. Practice

Exercise. B1

If $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$, show that $E(\hat{\boldsymbol{\varepsilon}}^\top \mathbf{A} \hat{\boldsymbol{\varepsilon}}) = \sigma^2 \text{tr}\{\mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M})\}$. For $n = 3$, $\mathbf{X} = \mathbf{1}$ and $v_{st} = \rho^{|s-t|}$, compute it and show that it decreases in ρ .

Exercise. B2

Suppose $\mathcal{C}(\mathbf{X})$ is spanned by the eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ of $\mathbf{A}$. Show that $\nu_k = \lambda_{k+p}$ for all k , so that $d = d_U$. Which columns of $\mathbf{X}$ have this property, and what do they look like?

Solution. Then $\mathcal{C}(\mathbf{X})^\perp$ is spanned by $\mathbf{v}_{p+1}, \dots, \mathbf{v}_n$, and $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp$ has eigenvalues $\lambda_{p+1}, \dots, \lambda_n$. The $\mathbf{v}_j$ are cosines of increasing frequency, starting with the constant $\mathbf{v}_1$; $\mathbf{v}_2$ is a single slow half-wave, close to a centred linear trend. Regressors that are smooth functions of time are close to this case, so d is close to d_U for trend regressions, and the upper bound is the relevant one.

Exercise. B3

Show that d is unchanged if $\mathbf{y}$ is replaced by $c\mathbf{y} + \mathbf{X}\mathbf{a}$ for any $c \neq 0$ and $\mathbf{a}$. Use this to explain why its null distribution cannot depend on $\boldsymbol{\beta}$ or σ , without appeal to normality.

21.5.5.3 C. Going deeper

Exercise. C1

For the intercept-only model, show that under the hypothesis $\text{Var } d = 4(n-2)/\{(n-1)(n+1)\}$ exactly, so that the standard deviation of d is close to $2/\sqrt{n}$. (Use [Theorem 21.5.3\(b\)](#), $\sum_j \lambda_j^2 = \text{tr}(\mathbf{A}^2)$, and the entries of $\mathbf{A}$.) Compare with the Okun regression of [Example \(The exact test for the Okun regression\)](#).

Solution. Here $m = n - 1$ and the ν_k are $\lambda_2, \dots, \lambda_n$, so $\sum_k \nu_k = \text{tr } \mathbf{A} = 2(n-1)$ and $\sum_k \nu_k^2 = \text{tr}(\mathbf{A}^2)$, the sum of the squared entries of $\mathbf{A}$: $1 + 4(n-2) + 1$ from the diagonal and $2(n-1)$ from the off-diagonal, in total $6n-8$. Then $m \sum_k \nu_k^2 - (\sum_k \nu_k)^2 = (n-1)(6n-8) - 4(n-1)^2 = 2(n-1)(n-2)$, and $\text{Var } d = 4(n-1)(n-2)/\{(n-1)^2(n+1)\} = 4(n-2)/\{(n-1)(n+1)\}$. For $n = 202$, $2/\sqrt{202} = 0.141$, close to the standard deviation 0.1399 computed for the Okun design, which has one more column.

Exercise. C2

In the model $y_t = \beta y_{t-1} + \varepsilon_t$ with independent errors, fitted by least squares on $t = 2, \dots, n$, argue that the residuals are less autocorrelated than the errors would be under an autocorrelated alternative, so that d is biased towards 2. Relate this to the requirement in [Theorem 21.5.3](#) that $\mathbf{X}$ be fixed.